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Temperature Characteristics of a *Y*-Cut *Z*-Propagation $LiNbO_3$ Light Modulator for Application to Polarimeters

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The temperature characteristics of a *Y*-cut *Z*-propagation $LiNbO_3$ crystal light modulator, with manufacturing errors, in the absence and presence of an electric field have been investigated by analyses and experiments. According to our analyses, when the *Z*-axis of the $LiNbO_3$ crystal is at an angle of $0.22°$ with respect to the normal of the input surface of the crystal, we found the theoretical fluctuation of the normalized output-light intensity with temperature to be less than $7.75 \times 10^{-6}/°C$. This magnitude is less than 1% of the theoretical intensity fluctuation of a conventional temperature-compensation $LiNbO_3$ light modulator. The measured temperature characteristics of a prototype of this modulator were $2 \times 10^{-4}/°C$ in the absence of an electric field (OFF state) and $2.8 \times 10^{-4}/°C$ in the presence of an external field (ON state). During a running test of longer than 8 hours at room temperature, the intensity fluctuation of this prototype was 0.01% in the OFF state, and 0.07% in the ON state.

Keywords: $LiNbO_3$ crystal, light modulator, electro-optic effect, birefringence, manufacturing error

1. Introduction

Ellipsometers have been applied to measure the refractive indices, extinction coefficients, and thicknesses of films, and polarimeters been used to measure polarization properties such as the birefringence and optical activity of materials. To remove the need for the mechanical operation of such optical equipment and to shorten the measurement time, a light modulator with high speed and high accuracy is necessary. Several types of modulator that can be applied for these measurements have been reported.

For instance, there are a photoelastic modulator (PEM) which used the PE effect of a quartz,[1,2] a nematic liquid crystal modulator whose birefringence is controlled with low voltage,[3] and a temperature-compensation light modulator consisting of dual $LiNbO_3$ (LN) crystals,[4] named a dual-crystal modulator. Although the PEM has an advantage of being suitable for image processing because of its large aperture, it simultaneously has the disadvantages of temperature dependence due to the birefringence of SiO_2, a limited operating band of resonance frequency, and a bulky and massive structure. The liquid crystal device is small, thin, lightweight, and controllable with a very low driving voltage but has a large temperature dependence, and its response speed is extremely low.

The dual-crystal modulator is small and has a high-speed response. Its major drawback is the large temperature dependence of the electro optic (EO) coefficients r_{13} and r_{33} of the LN crystal used for the operation of the dual-crystal modulator. Another problem that affects the temperature dependence of this device is that it is impossible to fabricate two crystals with perfectly identical crystal length and to exactly match the optical axes of the front and rear crystals; intrinsic birefringence is therefore unavoidable.

To reduce the temperature dependence, which is a disadvantage common to each of the above devices, we have developed a *Y*-cut and *Z*-propagation LN light modulator (hereafter called a *Y*–*Z* modulator) and utilized it in a birefringence measurement system.[5] This *Y*–*Z* modulator can be applied to polarimeters for the following reasons. When linearly polarized light with a polarization at angle of 45° with respect to the *X*-axis of the LN crystal propagates through the crystal along the *Z*-axis in the presence of an electric field parallel to the *Y*-axis, plane-polarized components of incident light that vibrate along *X* or *Y* direction of the LN crystal are subjected to the EO effect with an equal magnitude. Since both components are sensitive to the ordinary refractive index of the LN crystal, the static retardation will theoretically be zero, and the temperature dependence of this device is extremely low. A notable difference in the case of the dual-crystal modulator is the temperature independence of the EO coefficient r_{22} of the *Y*–*Z* LN crystal which may reduce the temperature dependence of the dynamic retardation induced by an external electric field.[6]

These properties of the *Y*–*Z* modulator are widely known, and several applications have been reported.[7–9] Takizawa *et al.* have developed an interferometric waveguide modulator having polarization-independence[8] and a TE/TM mode splitter consisting of asymmetric *X*-branch waveguides[9] using *Y*–*Z* LN crystals. However, the temperature characteristics of the *Y*–*Z* modulator in the presence of an applied electric field and in the case of having manufacturing errors are still not known. In this study, the temperature characteristics of the *Y*–*Z* modulator with the manufacturing error that the directions of the principal axes of the LN crystal are inconsistent with those of the normal axes of the crystal surface are analyzed in detail, and the relation between the temperature and the modulation characteristics

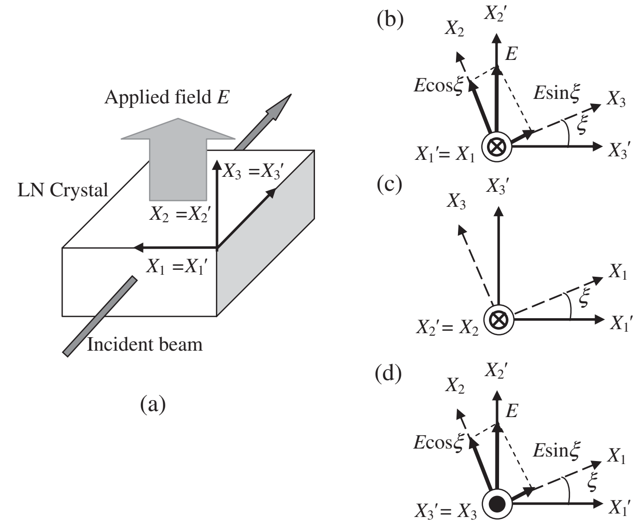

Fig. 1. Y–Z modulator (a) in the ideal case: the normal axes X_1', X_2', and X_3' and principal dielectric axes X_1, X_2, and X_3 coincide; (b) in Case I, the normal axis X_1' is fixed to the principal axis X_1, and the X_2' and X_3' axes are rotated by a small angle ξ relative to the X_2 and X_3 axes, respectively; (c) in Case II, X_2' is fixed to X_2, and X_1' and X_3' are rotated by an angle ξ relative to X_1 and X_3, respectively; (d) in Case III, X_3' is fixed to X_3, and X_1' and X_2' are rotated by an angle ξ relative to X_1 and X_2, respectively.

of the Y–Z modulator is experimentally obtained. Furthermore, the relation between the temperature and the performance of the dual-crystal modulator is also analyzed and compared with that of the Y–Z modulator.

2. Manufacturing Errors of the *Y–Z* Modulator

As illustrated in Fig. 1, X_1 ($= X$), X_2 ($= Y$), and X_3 ($= Z$) are the principal axes of the crystal, while X_1', X_2', and X_3' are the normal lines perpendicular to the crystal surface. We define the new rectangular coordinate system as X_1', X_2', and X_3' and the old rectangular coordinate system as X_1, X_2, and X_3. Opposite faces of the crystal are assumed to be perfectly parallel. In the case of no manufacturing error, the new axes and the old axes completely agree [Fig. 1(a)]. In this case, the Y–Z modulator has no static retardation because the direction of beam propagation agrees with the X_3 axis of the crystal. When light with linear polarization at an angle of 45° with respect to X_1 is incident on the LN crystal, the two plane-polarized components, the vibrations of which are parallel to the X_1 and X_2 axes, are sensitive to the ordinary index of refraction of the crystal and the retardation will be zero. In practical crystal processing, it is almost impossible to achieve $X_i = X_{j'}$ $(i, j = 1, 2, 3)$; there is often an angle error of 0.5 to 5 min. between X_i and $X_{j'}$. Here, we analyze this manufacturing error, which affects the retardation and thus the intensity of the modulated output light.

Manufacturing errors generally occur simultaneously in three dimensions, the analysis of which requires extremely complicated calculus. For simplification, we performed the analysis in three stages, and in each stage we fixed one of the three new axes (X_1', X_2', and X_3') of the crystal surfaces to be parallel to the corresponding old axis (X_1, or X_2, or X_3) of the crystal and assumed that the other new axes are rotated by a small angle of ξ about the fixed old axis. For each case, we calculated the static retardation due to natural birefringence as well as the dynamic retardation induced by an external electric field.

2.1 *In the case that the X_1' axis is parallel to the X_1 axis (Case I)*

Between the old and new coordinate systems, the following equation holds:

$$X_i = a_{ji} X'_j, \quad i, j = 1, 2, 3 \tag{1}$$

where a_{ji} is the direction cosine of X'_j with respect to X_i.

First, let us discuss the case that the X_1' axis is fixed to the principle dielectric axis X_1 of the LN crystal, and the X_2' and X_3' axes are rotated by a small angle of ξ relative to the X_2 and X_3 axes, respectively, as shown in Fig. 1(b). In this case, eq. (1) is rewritten as

$$\begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\xi & -\sin\xi \\ 0 & \sin\xi & \cos\xi \end{pmatrix} \begin{pmatrix} X'_1 \\ X'_2 \\ X'_3 \end{pmatrix}. \tag{2}$$

Owing to the above-mentioned manufacturing error, in the presence of an electric field applied parallel to X_2', the change in refractive index induced by the EO effect is

$$\begin{pmatrix} 0 & -r_{22} & r_{13} \\ 0 & r_{22} & r_{13} \\ 0 & 0 & r_{33} \\ 0 & r_{51} & 0 \\ r_{51} & 0 & 0 \\ -r_{22} & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ E\cos\xi \\ E\sin\xi \end{pmatrix} = \begin{pmatrix} -r_{22}E\cos\xi + r_{13}E\sin\xi \\ r_{22}E\cos\xi + r_{13}E\sin\xi \\ r_{33}E\sin\xi \\ r_{51}E\cos\xi \\ 0 \\ 0 \end{pmatrix}. \tag{3}$$

Substituting eq. (3) into the equation for the indicatrix, we obtain the following equation for the electrooptically perturbed indicatrix:

$$\begin{aligned} &\left(\frac{1}{n_o^2} - r_{22}E\cos\xi + r_{13}E\sin\xi\right)X_1^2 + \left(\frac{1}{n_o^2} + r_{22}E\cos\xi + r_{13}E\sin\xi\right)X_2^2 \\ &\quad + \left(\frac{1}{n_e^2} + r_{33}E\sin\xi\right)X_3^2 + 2r_{51}E\cos\xi X_2X_3 = 1, \end{aligned} \tag{4}$$

where r_{ij} is the EO coefficient of the LN crystal, E is the external electric field, and n_o and n_e are the ordinary and extraordinary refractive indices, respectively.

Substituting eq. (2) into eq. (4), we obtain the following equation:

$$\begin{aligned}
&\left[\frac{1}{n_o^2} - E(r_{22}\cos\xi - r_{13}\sin\xi)\right]X_1'^2 \\
&+\left[\frac{\cos^2\xi}{n_o^2} + \frac{\sin^2\xi}{n_e^2} + E(r_{22}\cos^3\xi + r_{13}\sin\xi\cos^2\xi + r_{33}\sin^3\xi + 2r_{51}\cos^2\xi\sin\xi)\right]X_2'^2 \\
&+\left[\frac{\sin^2\xi}{n_o^2} + \frac{\cos^2\xi}{n_e^2} + E(r_{22}\sin^2\xi\cos\xi + r_{13}\sin^3\xi + r_{33}\sin\xi\cos^2\xi - 2r_{51}\sin\xi\cos^2\xi)\right]X_3'^2 \\
&+2\cos\xi\left[\left(\frac{1}{n_e^2} - \frac{1}{n_o^2}\right)\sin\xi - E\sin\xi(r_{22}\cos\xi + r_{13}\sin\xi - r_{33}\sin\xi) + r_{51}E(\cos^2\xi - \sin^2\xi)\right]X_2'X_3' = 1.
\end{aligned} \tag{5}$$

Because light propagates along $X_3{}'$, eq. (5) is simplified by setting $X_3{}' = 0$ and becomes

$$\begin{aligned}
&\left[\frac{1}{n_o^2} - E(r_{22}\cos\xi - r_{13}\sin\xi)\right]X_1'^2 \\
&+\left[\frac{\cos^2\xi}{n_o^2} + \frac{\sin^2\xi}{n_e^2} + E(r_{22}\cos^3\xi + r_{13}\sin\xi\cos^2\xi + r_{33}\sin^3\xi + 2r_{51}\sin\xi\cos^2\xi)\right]X_2'^2 = 1.
\end{aligned} \tag{6}$$

The new refractive index $n_{x1'}$ for a wave polarized along the $X_1{}'$ axis is expressed as

$$n_{x1'} = n_o + \frac{1}{2}n_o^3(r_{22}\cos\xi - r_{13}\sin\xi)E, \tag{7}$$

and, for a wave polarized along $X_2{}'$ the new refractive index $n_{x2'}$ is given by

$$n_{x2'} = n_{oe} - \frac{1}{2}n_{oe}^3(r_{22}\cos^3\xi + r_{13}\sin\xi\cos^2\xi + r_{33}\sin^3\xi + 2r_{51}\sin\xi\cos^2\xi)E, \tag{8}$$

where

$$\frac{1}{n_{oe}^2} = \frac{\cos^2\xi}{n_o^2} + \frac{\sin^2\xi}{n_e^2}. \tag{9}$$

In the presence of the external electric field, the EO crystal exhibits the inverse piezoelectric effect as well as the EO effect. Strain caused by the inverse piezoelectric effect changes the crystal length and results in the phase change of light propagating through the crystal. As eq. (1) holds between the old and new coordinate systems, the relation between the strain $S_{ij}{}'$ for the new coordinates and the strain S_{kl} for the old coordinates is given by

$$S_{ij}{}' = a_{ik}a_{jl}S_{kl}. \tag{10}$$

On the old coordinates, the strain induced by the applied electric field E along $X_2{}'$ is given by

$$\begin{aligned}
&(S_1 \quad S_2 \quad S_3 \quad S_4 \quad S_5 \quad S_6) \\
&= (0 \quad E\cos\xi \quad E\sin\xi)\begin{pmatrix} 0 & 0 & 0 & 0 & d_{15} & -2d_{22} \\ -d_{22} & d_{22} & 0 & d_{15} & 0 & 0 \\ d_{31} & d_{31} & d_{33} & 0 & 0 & 0 \end{pmatrix} \\
&= (-d_{22}E\cos\xi + d_{31}E\sin\xi \quad d_{22}E\cos\xi + d_{31}E\sin\xi \quad d_{33}E\sin\xi \quad d_{15}E\cos\xi \quad 0 \quad 0),
\end{aligned} \tag{11}$$

where $S_1 = S_{11}$, $S_2 = S_{22}$, $S_3 = S_{33}$, $S_4 = S_{23} + S_{32}$, $S_5 = S_{31} + S_{13}$, and $S_6 = S_{12} + S_{12}$, and d_{ij} is the piezoelectric constant of the LN crystal.

Since the beam propagates along $X_3{}'$, here we consider only the change in the crystal length in this direction. From eqs. (10) and (11), the strain along $X_3{}'$ becomes

$$S_3' = (d_{22}\sin^2\xi\cos\xi + d_{31}\sin^3\xi + d_{33}\sin\xi\cos^2\xi - d_{15}\sin\xi\cos^2\xi)E, \tag{12}$$

and the change of the crystal length in the same direction will be

$$\Delta L_{x3'} = S_3'L, \tag{13}$$

where L is the length of the crystal in the absence of the applied field.

The retardation θ, with the external electric field applied along $X_2{}'$, is given by

$$\theta = \frac{2\pi}{\lambda}(n_{x2'} - n_{x1'})(L_{x3'} + \Delta L_{x3'}) = \theta_s + \theta_d, \tag{14}$$

where θ_s is the natural or static retardation due to the EO crystal without the external electric field and θ_d is the dynamic retardation induced by the external field, which expressed by

$$\theta_s = \frac{2\pi}{\lambda}(n_o - n_{oe})L, \tag{15}$$

$$\theta_d = \frac{\pi}{\lambda} n_o^3 EL \left[\begin{array}{l} r_{22}\cos\xi - r_{13}\sin\xi + \left(\frac{n_{oe}}{n_o}\right)^3 \left(r_{22}\cos^3\xi + r_{13}\sin\xi\cos^2\xi + r_{33}\sin^3\xi + 2r_{51}\sin\xi\cos^2\xi\right) \\ + \frac{2(n_o - n_{oe})}{n_o^3}\left(d_{22}\sin^2\xi\cos\xi + d_{31}\sin^3\xi + d_{33}\sin\xi\cos^2\xi - d_{15}\sin\xi\cos^2\xi\right) \end{array} \right]. \tag{16}$$

2.2 *In the case that the $X_2{}'$ axis is parallel to X_2 (Case II)*

When $X_2{}'$ is fixed to the principle axis X_2 of the LN crystal, and $X_1{}'$ and $X_3{}'$ are rotated by a small angle of ξ relative to X_1 and X_3, respectively as shown in Fig. 1(c), in the presence of an external field applied parallel to $X_2{}'$, the change in refractive index induced by the EO effect and the electrooptically perturbed indicatrix are expressed as

$$\begin{pmatrix} 0 & -r_{22} & r_{13} \\ 0 & r_{22} & r_{13} \\ 0 & 0 & r_{33} \\ 0 & r_{51} & 0 \\ r_{51} & 0 & 0 \\ -r_{22} & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ E \\ 0 \end{pmatrix} = \begin{pmatrix} -r_{22}E \\ r_{22}E \\ 0 \\ r_{51}E \\ 0 \\ 0 \end{pmatrix}, \tag{17}$$

$$\left(\frac{1}{n_o^2} - r_{22}E\right)X_1^2 + \left(\frac{1}{n_o^2} + r_{22}E\right)X_2^2 + \frac{x_3^2}{n_e^2} + 2r_{51}EX_1X_3 = 1. \tag{18}$$

Referring to Fig. 1(c), eq. (1) is rewritten as

$$\begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} \cos\xi & 0 & \sin\xi \\ 0 & 1 & 0 \\ -\sin\xi & 0 & \cos\xi \end{pmatrix} \begin{pmatrix} X_1{}' \\ X_2{}' \\ X_3{}' \end{pmatrix}. \tag{19}$$

Substituting eq. (19) into eq. (18) and carrying out a mathematical argument similar to eqs. (5)–(9), we obtain the refractive indices $n_{x1'}$ and $n_{x2'}$ for light polarized parallel to $X_1{}'$ and $X_2{}'$, respectively, as follows:

$$n_{x1'} = n_{oe} + \frac{1}{2} n_{oe}^3 (r_{22}\cos^2\xi + r_{51}\sin 2\xi)E, \tag{20}$$

$$n_{x2'} = n_o - \frac{1}{2} n_o^3 r_{22} E. \tag{21}$$

The strain owing to the presence of the external electric field applied along $X_2{}'$ is expressed by

$$\begin{aligned} &(S_1 \quad S_2 \quad S_3 \quad S_4 \quad S_5 \quad S_6) \\ &= (0 \quad E \quad 0) \begin{pmatrix} 0 & 0 & 0 & 0 & d_{15} & -2d_{22} \\ -d_{22} & d_{22} & 0 & d_{15} & 0 & 0 \\ d_{31} & d_{31} & d_{33} & 0 & 0 & 0 \end{pmatrix} \\ &= (-d_{22}E \quad d_{22}E \quad 0 \quad d_{15}E \quad 0 \quad 0). \end{aligned} \tag{22}$$

From eqs. (10) and (22), the strain along $X_3{}'$ is given by

$$S_3' = -d_{22}E\sin^2\xi, \tag{23}$$

which leads to a change in the crystal length in the same direction of

$$\Delta L = S_3' L = -d_{22}EL\sin^2\xi. \tag{24}$$

Substituting eqs. (20), (21), and (24) into eq. (14) yields the following static and dynamic retardations θ_s and θ_d, respectively:

$$\theta_s = \frac{2\pi}{\lambda}(n_o - n_{oe})L, \tag{25}$$

$$\theta_d = -\frac{\pi}{\lambda} n_o^3 EL \left\{ r_{22}\left[1 + \left(\frac{n_{oe}}{n_o}\right)^3 \cos^2\xi\right] + \left(\frac{n_{oe}}{n_o}\right)^3 r_{51}\sin 2\xi - \frac{2(n_o - n_{oe})}{n_o^3} d_{22}\sin^2\xi \right\}. \tag{26}$$

2.3 *In the case that the $X_3{}'$ axis is parallel to X_3 (Case III)*

When $X_3{}'$ is fixed to the principle dielectric axis X_3 of the LN crystal, a beam propagating along $X_3{}'$ undergoes zero birefringence, even $X_1{}'$ and $X_2{}'$ are rotated by any angle ξ relative to X_1 and X_2, respectively, as shown in Fig. 1(d), in the presence of an electric field applied parallel to $X_2{}'$, the rotation of the principal axes does not occur. Thus, the refractive indices $n_{x1'}$ and $n_{x2'}$ on $X_1{}'$ and $X_2{}'$, respectively, are expressed as

$$n_{x1'} = n_o + \frac{1}{2} n_o^3 r_{22} E, \tag{27}$$

$$n_{x2'} = n_o - \frac{1}{2} n_o^3 r_{22} E. \tag{28}$$

Since the strain along $X_3{}'$ is zero, there is no change in the crystal length in the $X_3{}'$ direction. The static and dynamic retardations θ_s and θ_d are given by

$$\theta_s = 0, \tag{29}$$

$$\theta_d = \frac{2\pi}{\lambda} n_o^3 r_{22} LE. \tag{30}$$

The retardation in Case III is the same as that of an ideal *Y–Z* modulator without a manufacturing error.

3. Temperature Dependence of Refractive Indices, EO Coefficients, and Lattice Constants of LN Crystal

To analyze the temperature characteristics of the *Y–Z* modulator, we require temperature dependence of the refractive indices, EO coefficients, and lattice constants of the LN crystal. The temperature and/or wavelength

Table 1. Coefficients in eq. (31) for ordinary and extraordinary refractive indices.[11)]

Coefficient	n_o	n_e
a_1	4.905	4.582
a_2	1.180×10^5	9.920×10^4
a_3	2.180×10^2	2.110×10^2
a_4	-2.720×10^{-8}	-2.190×10^{-8}
b_1	2.230×10^{-2}	5.270×10^{-2}
b_2	-2.970×10^{-5}	-4.910×10^{-5}
b_3	2.140×10^{-8}	2.300×10^{-8}

dependence of the refractive indices of the congruent LN crystal have been comprehensively investigated by several researchers. Nelson and Mikulyak[10)] measured the refractive indices of LN crystal at 24.5 °C over the wavelength range of 404.63–3051.48 nm. The data were confirmed by Smith *et al.*[11)] Furthermore, they measured the temperature-dependence of the refractive indices $n(T, \lambda)$ of LN crystal from 20 to 600 °C at two wavelengths (632.8 and 3392.2 nm) and obtained the following Sellmeier-type equation:[11)]

$$n^2(\lambda, T) = a_1 + \frac{a_2 + b_1 f(T)}{\lambda^2 - [a_3 + b_2 f(T)]^2} + b_3 f(T) + a_4\lambda^4, \tag{31}$$

$$f(T) = (T - 24.5)(T + 546), \tag{32}$$

where T is the temperature in °C and λ is the wavelength in nm. Table 1 lists the coefficients in eq. (31) for the ordinary and extraordinary refractive indices of the congruent LN crystal.

The temperature-dependence of EO coefficients r_{13}, r_{33}, and r_{22} of the congruent LN crystal has been measured by Zook *et al.* at temperatures from 20 to 200 °C.[6)] Through their measurement of the relations between half-wave voltage and temperature, they found that r_{13} and r_{33} linearly decrease with increasing temperature, and that r_{22} remains constant up to 125 °C. According to their measurement results, the temperature gradients of coefficients r_{13}, r_{33}, and r_{22} are as follows:

$$\begin{aligned}\frac{dr_{13}}{dT} &= -0.0046\,\text{pm/V°C}\\ \frac{dr_{33}}{dT} &= -0.016\,\text{pm/V°C}\\ \frac{dr_{22}}{dT} &\approx 0\,\text{pm/V°C}.\end{aligned} \tag{33}$$

We also measured the temperature-dependence of r_{22} for the congruent LN crystal and obtained the same results as those of Zook *et al.* (see Fig. 2). In §5, we describe this experiment in detail.

The temperature-dependence of the lattice constants of the congruent LN crystal has been measured by Gallagher *et al.*[12)] According to their paper, the temperature dependence of the lattice constant D of the material is given by

$$D = a(1 + bT + cT^2 + dT^3)\ (T \text{ in °C}). \tag{34}$$

Table 2 lists the coefficients in the above equation.

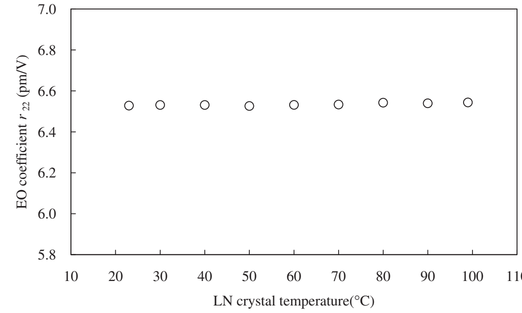

Fig. 2. Relation between EO coefficient r_{22} of LN crystal and temperature.

Table 2. Coefficients in eq. (34).[12)]

Temperature range (K)	a	b ($\times 10^{-6}$)	c ($\times 10^{-6}$)	d ($\times 10^{-6}$)
X_1-axis				
60–220	5.149	13	0	−98.1
220–523	5.149	13.43	17.55	−16.3
523–1373	5.1485	16.29	4.83	0
X_3-axis				
60–373	13.8673	4.09	4.4	−20.2
373–1373	13.8633	4.4	0	−3.69

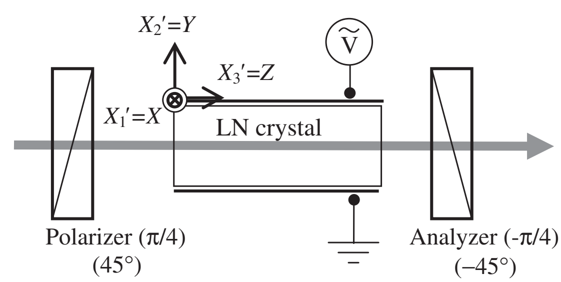

Fig. 3. Optical arrangement used to measure the temperature-dependence of the Y–Z modulator.

3.1 Temperature-dependence of the Y–Z modulator

Figure 3 shows the optical arrangement used to measure the temperature dependence of the Y–Z modulator. The modulator is inserted between two crossed polarizers, the transmission axes of which are at 45 and 135° with respect to the X_1'-axis. The normalized intensity of light emerging from the analyzer is expressed as

$$I = \frac{1 - \cos(\theta_s + \theta_d)}{2}. \tag{35}$$

Substituting eqs. (31)–(34) into eqs. (15), (16), (25), (26), (29), and (30), and then substituting the static and dynamic retardations θ_s and θ_d, into eq. (35), we obtain the temperature-dependence of the Y–Z modulator with the manufacturing error discussed in §2. Figures 4–6 illustrate the calculated results for the Y–Z modulator with a crystal length of $L = 20.6$ mm. In this analysis, the wavelength of the incident beam is assumed to be $\lambda = 632.8$ nm, the

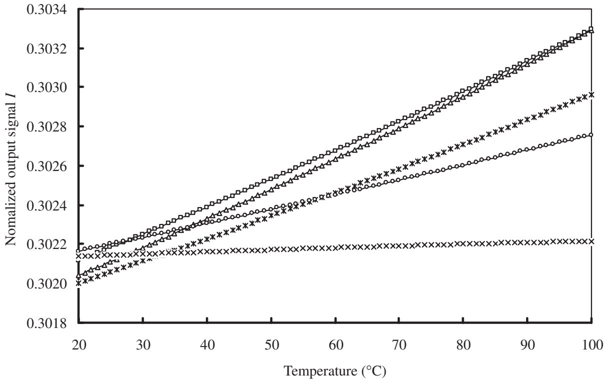

Fig. 4. Dependence of output intensity on the temperature for the Y–Z modulator with a manufacturing error in Case I. Triangles: $\xi = 0.00^\circ$, squares: $\xi = 0.05^\circ$, asterisks: $\xi = 0.10^\circ$, circles: $\xi = 0.15^\circ$, crosses: $\xi = 0.20^\circ$.

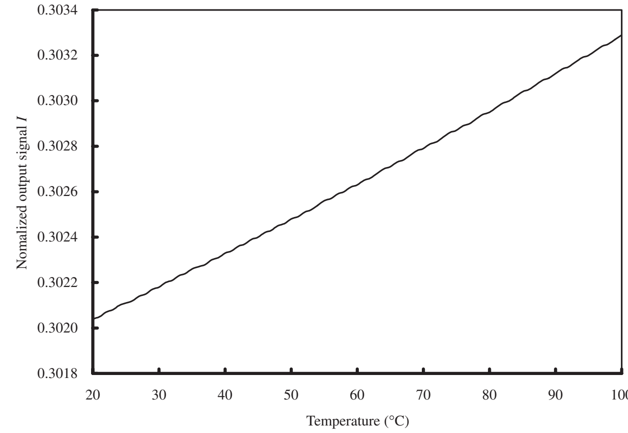

Fig. 6. Dependence of output intensity on the temperature for the Y–Z modulator with a manufacturing error in Case III.

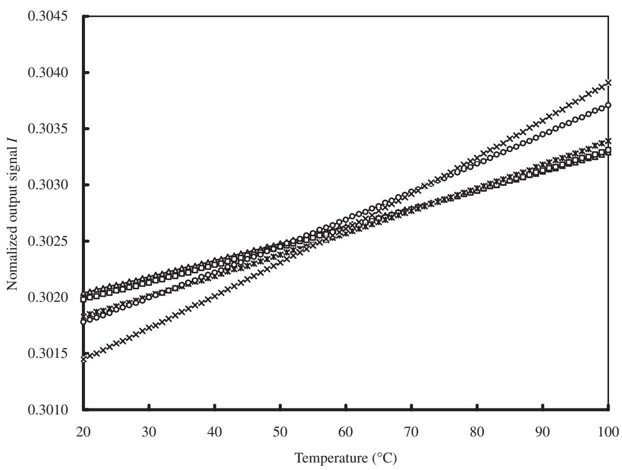

Fig. 5. Dependence of output intensity on the temperature for the Y–Z modulator with a manufacturing error in Case II. Triangles: $\xi = 0.00^\circ$, squares: $\xi = 0.05^\circ$, asterisks: $\xi = 0.10^\circ$, circles: $\xi = 0.15^\circ$, crosses: $\xi = 0.20^\circ$.

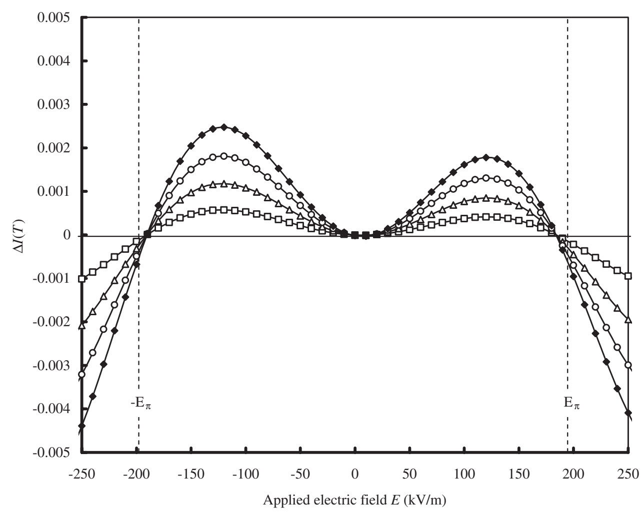

Fig. 7. $\Delta I(T)$–E characteristics of the Y–Z modulator with error $\xi = 0.1^\circ$ in Case I. Solid line: $T = 20\,^\circ$C, squares: $T = 40\,^\circ$C, triangles: $T = 60\,^\circ$C, circles: $T = 80\,^\circ$C, closed diamonds: $T = 100\,^\circ$C.

applied electric field is $E = 70\,\mathrm{kV/m}$, and the manufacturing error is $\xi = 0$, 0.05, 0.1, 0.15, or 0.2°. From Figs. 4–6, for the Y–Z modulator with a manufacturing error of $\xi < 0.2^\circ$, the fluctuation of the normalized output increases from 5×10^{-5} to $10^{-4}/^\circ$C when the temperature increases from 20 to 100 °C. From Fig. 4, the modulator with a manufacturing error of $\xi = 0.2^\circ$ is less temperature dependent than that with no error ($\xi = 0^\circ$). However, this is a phenomenon occurring in only a limited temperature range. The output intensity will be changed by the external electric field therefore, it is necessary to analyze the relation between intensity I and applied field E at various temperatures.

To obtain the I–E characteristics of the Y–Z modulator, we analyze the relation between E and the intensity fluctuation $\Delta I(T)$, which is the difference between the output intensity $I(T)$ at temperature T°C and the standard intensity $I(20)$ at 20 °C and is expressed as

$$\Delta I(T) = I(T) - I(20). \tag{36}$$

Figures 7–9 show the $\Delta I(T)$–E characteristics of the modulator when one of the normal axes $X_1{}'$, $X_2{}'$, or $X_3{}'$ is parallel to the corresponding crystal axis X_1, X_2, or X_3, and the other two axes are rotated about the fixed axis by $\xi = 0.1^\circ$. In this analysis, the crystal length L is taken as 20.6 mm, the wavelength of incident light is $\lambda = 632.8$ nm, and the crystal temperature is $T = 40$, 60, 80, or 100 °C. From Figs. 7–9, the following results as below are apparent:

1. $\Delta I(T)$ increases with increasing the temperature T.
2. $\Delta I(T)$ increases with increasing the applied field E.
3. Taking E_π as the half-wave electric field (= 196.5 kV/m) of this modulator, $\Delta I(T)$ becomes zero when the applied field approaches $E \approx mE_\pi$ ($m = 0, \pm 1, \pm 2, \ldots$).
4. When $E < |E_\pi|$, and the working temperature changes from 20 to 100 °C, the maximum fluctuation of the normalized intensity is $2.63 \times 10^{-5}/^\circ$C for the device in Case III and $3.13 \times 10^{-5}/^\circ$C in Cases I and II.

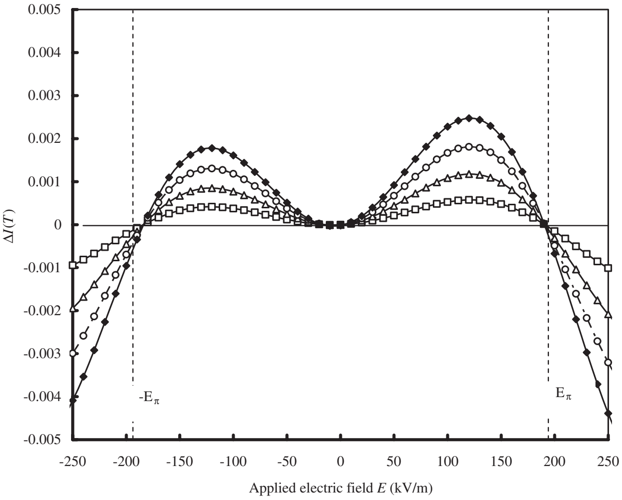

Fig. 8. $\Delta I(T)$–E characteristics of the Y–Z modulator with error $\xi = 0.1°$ in Case II. Solid line: $T = 20\,°C$, squares: $T = 40\,°C$, triangles: $T = 60\,°C$, circles: $T = 80\,°C$, closed diamonds: $T = 100\,°C$.

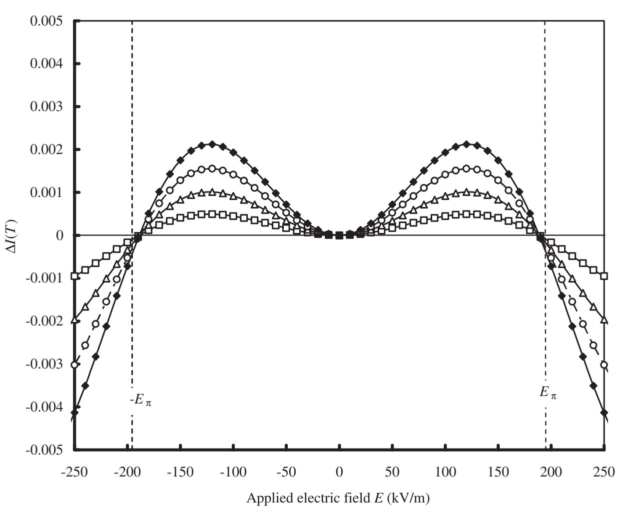

Fig. 9. $\Delta I(T)$–E characteristics of the Y–Z modulator with error $\xi = 0.1°$ in Case III. Solid line: $T = 20\,°C$, squares: $T = 40\,°C$, triangles: $T = 60\,°C$, circles: $T = 80\,°C$, closed diamonds: $T = 100\,°C$.

The fourth result demonstrates the excellent temperature characteristic of the Y–Z modulator. Owing to the third result, when the modulator is applied to on–off optical switches such as directional couplers,[13] EO branching waveguide switches,[14] waveguide switches using total internal reflection[15] and Y-junction optical switches,[16] their temperature characteristics are expected markedly improved.

Figure 10 shows the $\Delta I(T)$–E characteristics of the Y–Z modulator in Case I when $\xi = 0.22°$. In the field range of -30 to $190\,\mathrm{kV/m}$, the maximum fluctuation of the normalized intensity is less than $7.75 \times 10^{-6}/°C$, which is approximately one-quarter of that of the modulator in Case III (see Fig. 9). Compared with the ideal device, the

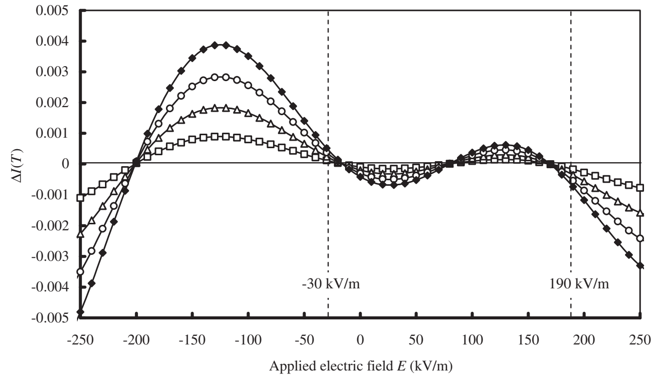

Fig. 10. $\Delta I(T)$–E characteristics of the Y–Z modulator with error $\xi = 0.22°$ in Case I. Solid line: $T = 20\,°C$, squares: $T = 40\,°C$, triangles: $T = 60\,°C$, circles: $T = 80\,°C$, closed diamonds: $T = 100\,°C$.

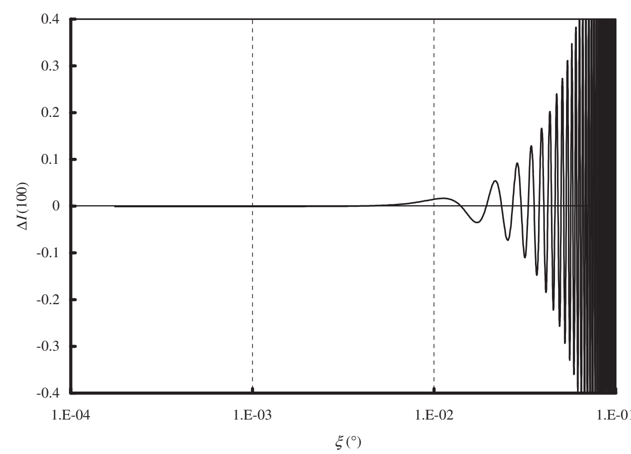

Fig. 11. Relation between $\Delta I(100)$ and the error angle ξ of the Y–Z modulator in Case I. The applied electric field E is $200\,\mathrm{kV/m}$.

device with some manufacturing errors thus exhibits decreased intensity fluctuation caused by the change in temperature. The result was the same for the device in Case II. As it is relatively easy to manufacture the Y–Z modulator with a rotation angle of $\xi = 0.2°$, the characteristics as shown in Fig. 10 can be readily realized.

Figure 11 shows the $\Delta I(100)$–E characteristics of the Y–Z modulator in Case I. In this figure, when $\xi < 0.38°$ and $E = 200\,\mathrm{kV/m}$, the maximum fluctuation of the optical intensity is controlled to less than $5 \times 10^{-5}/°C$ in the temperature range of 20–100 °C.

4. Temperature Characteristic of the Dual-Crystal Modulator

Figure 12 shows the typical arrangement of a dual-crystal modulator. It consists of two LN crystals sandwiching a half-wave plate, the fast axis of which is at an angle of 45° with respect to the principal axis X_3 of each LN crystal. Owing to this arrangement, the extraordinary ray in the first crystal becomes the ordinary ray in the second crystal and *vice versa*. Therefore, if the size of the two crystals is

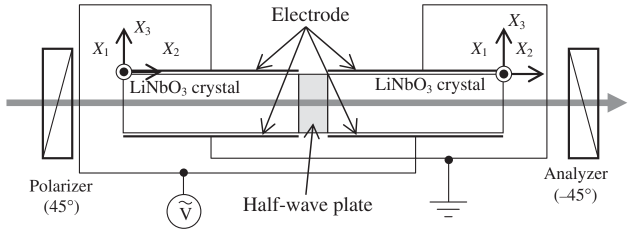

Fig. 12. Configuration of the dual-crystal modulator.

exactly the same and there is no manufacturing error and no assembly error, there will be no the temperature dependence in the absence of an electric field, owing to the zero natural birefringence. However, the electrooptically induced retardation of the ideal dual-crystal modulator changes markedly with changing temperature because of the large dependence of the EO coefficients r_{13} and r_{33} on temperature. Furthermore, this device will have various processing errors because the arrangement of the dual-crystal modulator is more complicated than that of the Y–Z modulator.

To simplify the analysis process, here we consider the case that the length of the two LN crystals is different. Taking the lengths of the first and second crystals as L, and $L + \Delta L$, the static retardation θ_s due to the natural birefringence and the dynamic retardation θ_d caused by the EO effect are, respectively,

$$\theta_s = \frac{2\pi}{\lambda}(n_o - n_e)\Delta L, \quad (37)$$

$$\theta_d = -\frac{\pi}{\lambda}E[(n_e{}^3 r_{33} - n_o{}^3 r_{13}) + (n_o - n_e)d_{31}](2L + \Delta L). \quad (38)$$

Substituting eqs. (37) and (38) into eqs. (35) and (36), we obtain the temperature characteristic of the dual-crystal modulator. Figure 13 shows the $\Delta I(T)$–E characteristics of this modulator with $\Delta L = 0$ and $L = 10.3$ mm at $\lambda = 632.8$ nm. Since $\Delta L = 0$, the output intensity remains constant with changing temperature when $E = 0$. The dynamic retardation θ_d includes temperature dependent parameters such as the refractive indices (n_o, n_e), EO coefficients (r_{13}, r_{33}), and crystal length L and has a clear temperature dependence when $E \neq 0$. The half-wave electric field E_π of the dual-crystal modulator is approximately 144.7 kV/m for $L = 10.3$ mm. The maximum fluctuation of the output intensity of this modulator operating at an applied field of between $-E_\pi$ to E_π, is about 6.60×10^{-4}/°C at temperatures 20–100 °C, as shown in Fig. 13. This value is about 25 times larger than that of the Y–Z modulator with no manufacturing error.

Next, let us discuss the case when the dual-crystal modulator has a manufacturing error of 0.01% ($\Delta L = 1.03\,\mu$m). The other parameters are assumed to be the same as those for the ideal case. Figure 14 shows the temperature characteristic of this modulator. The maximum fluctuation of the output intensity of this modulator operating at an applied field of between $-E_\pi$ to E_π, is about 9.92×10^{-4}/°C at temperatures of 20–100 °C. This value is about 32 times larger than that of the Y–Z modulator with a manufacturing error of $\xi = 0.1°$. Figure 14 shows that the intensity of the output light of the dual-crystal modulator with a manufacturing error of ΔL is affected by influence of temperature even in the case of $E = 0$. The comparison between the temperature characteristic of the dual-crystal modulator with a manufacturing error of 0.01% and that of the Y–Z modulator with a manufacturing error of $\xi = 0.22°$ (Fig. 10) shows that the maximum fluctuation of the output intensity of the former device is about 127 times larger than that of the latter device.

The conclusion of this section is as follows: the dual-crystal modulator has the advantage that the half-wave electric field E_π is less than that of the Y–Z modulator but has the disadvantage that the maximum fluctuation of the output intensity at each temperature is 25–127 times larger than that of the Y–Z modulator.

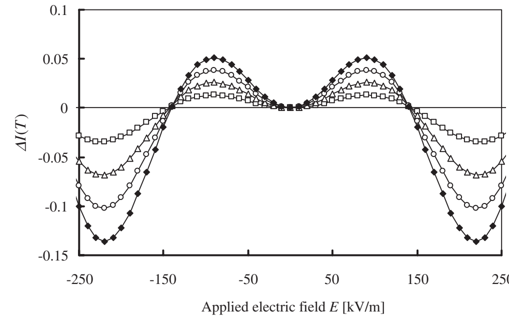

Fig. 13. Relation between $\Delta I(T)$ and the external field E of the dual-crystal modulator with $\Delta L = 0$. Solid line: $T = 20\,°$C, squares: $T = 40\,°$C, triangles: $T = 60\,°$C, circles: $T = 80\,°$C, closed diamonds: $T = 100\,°$C.

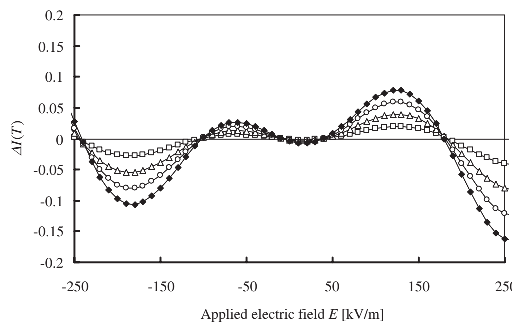

Fig. 14. $\Delta I(T)$–E characteristics of the dual-crystal modulator with error $\Delta L = 1.03\,\mu$m. Solid line: $T = 20\,°$C, squares: $T = 40\,°$C, triangles: $T = 60\,°$C, circles: $T = 80\,°$C, closed diamonds: $T = 100\,°$C.

5. Fabrication of a Y–Z Modulator and Measurement of Its Temperature Characteristic

In the preceding sections, we analyzed the temperature characteristic of the Y–Z modulator. In this section we discuss the measurement of the temperature characteristic of a prototype modulator.

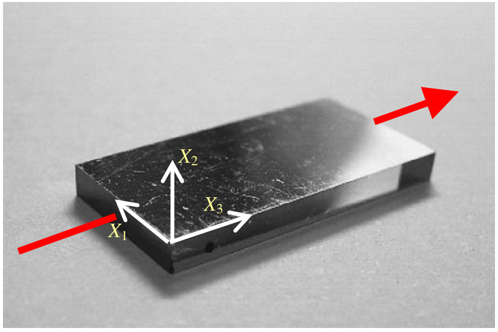

Fig. 15. (Color online) Prototype Y–Z modulator.

Table 3. Specifications of the prototype Y–Z modulator.

Dimensions (mm)		
Along X-axis	Along Y-axis	Along Z-axis
2.00 ± 0.01	9.93 ± 0.02	20.01 ± 0.01
Axis quality (°)		
X-axis	Y-axis	Z-axis
$< \pm 0.1$	$< \pm 0.01$	$< \pm 0.1$
Parallelism of [010] surfaces (°)		
0.01		

5.1 *Fabrication of a prototype Y–Z modulator and a temperature controller*

The prototype consists of a Y-cut and Z-propagation congruent LN crystal with a manufacturing error of $\xi < 0.1°$. The surfaces of the (001) crystal plane were polished and covered with a Cr–Ni–Au alloy film electrode (thickness = 200 nm). The profile of the prototype and its specifications are shown in Fig. 15 and Table 3, respectively. The crystal length was measured using a video measuring system (Nikon VM-150N) and the crystal axes were measured using a piezo-goniometer (Rigaku 2991F2).

To measure the temperature characteristic of this prototype, we fabricated a small temperature controller, as shown in Figs. 16 and 17. This controller consists of a temperature control unit, a Peltier device driver, thermometers, and temperature sensors. Two sides of the temperature control unit are opened for the input and output light beams therefore, the temperature distribution of the whole unit is not uniform. This problem is eliminated by installing a small modulator with dimensions given shown in Table 3 in the center of the control unit (sectional area = $120 \times 120\,\mathrm{mm}^2$). Type K thermocouples are attached to the upper and lower electrodes of the temperature control unit. This electrode and Peltier device are separated by an insulator. By controlling the current used to drive the Peltier device, which is determined by the temperature difference between the setting temperature established by the temperature controller and the electrode temperature detected by the thermocouple, we maintained the modulator temperature the preset temperature. The temperature is controlled from room temperature to 100 °C with resolution ±0.1 °C.

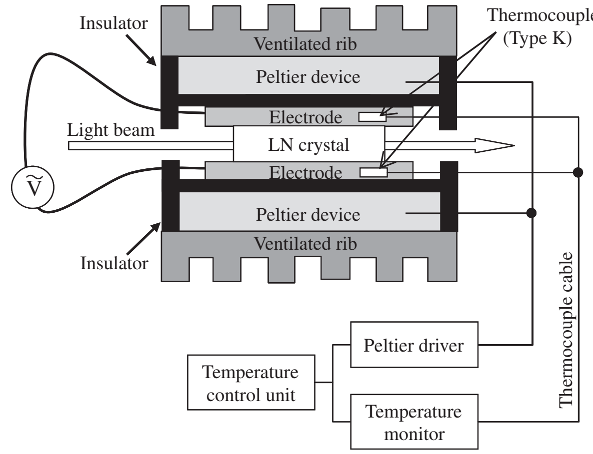

Fig. 16. Arrangement of the constant-temperature unit.

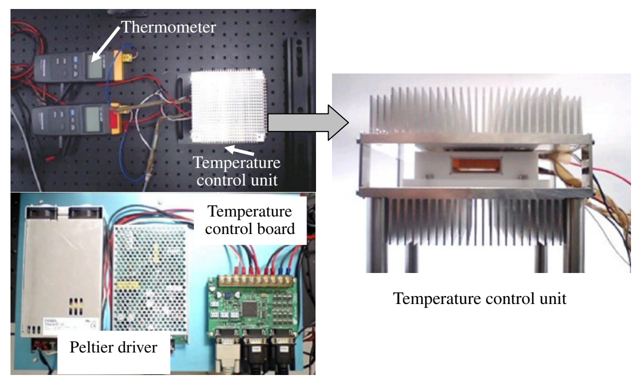

Fig. 17. (Color online) Temperature control unit.

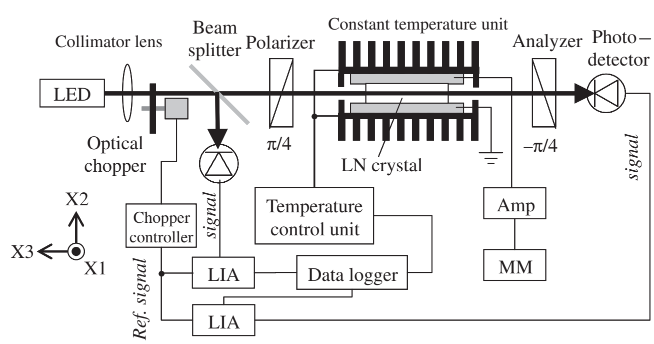

Fig. 18. Experimental setup. LIA: lock-in amplifier, MM: multimeter.

5.2 *Experiment and results*

Figure 18 shows the experimental setup for measurement of the relation between the temperature and output intensity signal of the prototype Y–Z modulator. Table 4 lists the principal instruments in the arrangement. To avoid multiple-reflection interference occurring inside the LN crystal, the light source employed a light-emitting diode (LED) with a central wavelength of 633 nm (a full width at half maximum of 10 nm). Part of the light from the LED and

Table 4. Principal instruments of the experimental setup.

Instrument	Manufacturer	Type	Description
Lock-in amplifier	NF	LI5640	Gain stability ± 100 ppm/°C
High voltage amplifier	Mastusada Precision	HEOPS-3B10	Output voltage stability $\pm 0.016\%$
Thermometer	LINE SEIKI	TC-850	Resolution 0.1 °C, Accuracy $\pm 0.2\%$
Photo-detector	Neoark	PA-241	Equivalent noise level 0.5 mVp–p/50 Ω
Function generator	Agilent Technol.	33120A	Output voltage 1 mV–10 V
Multi-meter	Agilent Technol.	34401A	Crest factor error $\pm 0.002\%$
Data logger	NEC/Avico	RA1300	Stored signal level 14 bit

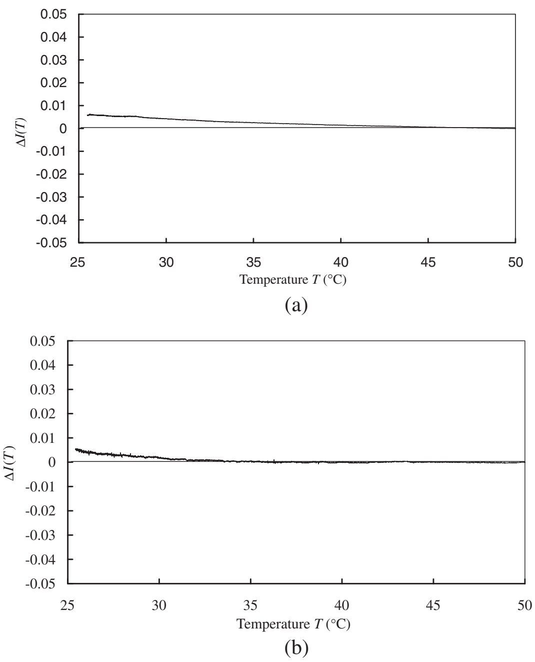

Fig. 19. Relation between $\Delta I(T)$ and T of the prototype Y–Z modulator in the (a) absence and (b) presence of the electric field. The light source used is an LED ($\lambda = 633$ nm).

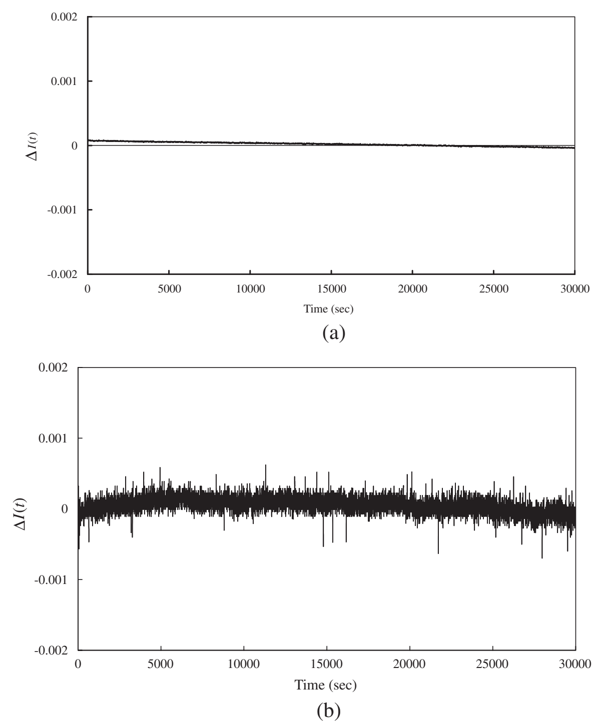

Fig. 20. Variation of optical intensity $\Delta I(t)$ of the prototype in the (a) absence and (b) presence (AC voltage 404 V) of electric field with time.

the intensity signal from the detector are recorded using a data logger to compensate for the fluctuation of the LED output.

The normalized output intensity change of the prototype Y–Z modulator when the crystal temperature changed from 50 to 25 °C is illustrated in Fig. 19. Figure 19(a) shows the measurement result in the absence of an applied voltage and Fig. 19(b) shows the result in the presence of a sinusoidal voltage (amplitude = 170.8 V, frequency 1 kHz).

The normalized intensity change $\Delta I(T)$ is used to compare of the output signal at crystal temperature T°C with that at 50 °C, and is given by

$$\Delta I(T) = \frac{I(T) - I(50)}{I(50)}. \tag{39}$$

The change of normalized output is about 2.8×10^{-4}/°C when the applied field is zero, and 2.0×10^{-4}/°C when an AC voltage is applied. Both results are larger than those obtained by simulation; however, $\Delta I(T)$ shows excellent stability superior to that calculated for the dual-crystal modulator with a very small manufacturing error of 0.01% shown in Fig. 13.

We also measured the variation of temporal output $\Delta I(t)$ of the prototype under the free condition using the temperature control unit. In this case $\Delta I(t)$ is given by

$$\Delta I(t) = \frac{I(t) - \bar{I}}{\bar{I}}, \tag{40}$$

where $I(t)$ is the intensity signal at time t, and $\bar{I}$ is the averaged intensity. The room temperature was controlled to 25 °C using only an air conditioner. Figure 20 shows the output intensity in the presence of AC voltages of 0 and 404 V. The signal fluctuation $\Delta I(t)$ of the prototype modulator in the absence of the applied voltage was less than 1×10^{-4}, while $\Delta I(t)$ in the presence of the applied AC voltage was less than 2×10^{-4} over a long time span and less than 7×10^{-4} over a short span. The most probable cause of the short span fluctuation is the physical phenomena related to the LN crystal, such as the electro optic effect, inverse piezoelectric effect, pyroelectric effect, photovoltaic effect, and photorefractive effect, because signal fluctuation was noticeable in the presence of the applied voltage. We measured the hourly fluctuation for the *Y–Z* modulator several times, but could not determine the factor causing the short span fluctuation.

6. Conclusions

The temperature characteristic of a *Y–Z* modulator utilized for polarimeters has been analyzed in detail, and its excellent stability has been experimentally verified. In our simulation, we assumed that one of the normals to the crystal surfaces is parallel to the principal dielectric axis of the crystal, and the other two normals are rotated by a small angle ξ about the fixed axis. Substituting the already known temperature characteristic of the refractive index, lattice constant, and electrooptic coefficient into the retardation equation obtained through the coordinate transformation, we simulated the relation between the modulated intensity signal, as a function of the temperature, and the manufacturing error. The properties of the *Y–Z* modulator (crystal length $= 20.6$ mm) are discussed in §3.1. These properties are markedly superior to those of the conventional temperature-compensation dual-crystal modulator.

To verify our simulation results, we fabricated a prototype using a congruent LN crystal and a temperature controller, and measured the output signal while changing the temperature. The variation of intensity signal of the prototype was 2.0×10^{-4}–2.8×10^{-4}/°C at temperatures from 20 to 50 °C. Since a conventional light modulator operates within the range from 0 to 90 °C, our next target is to confirm the high stability of the *Y–Z* modulator in this the temperature range. In an 8 hours running test at room temperature, variance of the output signal is 0.07% with an applied field on, and 0.01% with field off.

By applying this excellent property of the *Y–Z* modulator to polarimeters and ellipsometers, we can realize a measurement system with high speed and high stability. In particular, when the applied field is equal to an integer multiple of the half-wave field, the device exhibits temperature independence; thus an extremely stable optical measurement system can be achieved.

There are many EO crystals without the natural birefringence of the *Y*-cut *Z*-transmission LN crystal, for instance, GaAs and InP crystals belonging to the symmetry group $\bar{4}3m$; $Bi_{12}SiO_{20}$ and $Bi_{12}GeO_{20}$ (23); and KH_2PO_4, KD_2PO_4, $(NH_4)H_2PO_4$ ($\bar{4}2m$). In our future work, we will analyze and measure the temperature characteristics of light modulation using these crystals.

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